Card Counting

Learning the Plus/Minus counting system.
Any method of card counting exists to accomplish three goals:

- Tell the player when to bet more.
- Tell the player when to deviate from basic strategy.
- Tell the player when to take insurance.
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There are numerous counting strategies and side counts strategies, the simple plus minus count system is the easiest to learn and use and is the one recommended for multi-deck games. In the Plus/Minus count system each card is assigned a value:

2	3	4	5	6	7	8	9	10	A
+1	+1	+1	+1	+1	0	0	0	-1	-1

The reason the 2 – 6 have positive values is because their removal from the deck is a good thing for the player. The dealer needs those small cards to make a good total when hitting stiff hands that the player doesn't have to hit.

The reason the tens and the aces have minus values is because their removal is bad for the player. The player gets paid 3 to 2 for a snapper and thus needs tens and aces in order to get them. The player also wants tens and aces for double downs. And the player wants tens in the deck so the dealer will draw them and bust his hand.

So, if the total of all exposed cards is a negative number, the remaining composition of the deck is favorable to the house. If the total of all exposed cards is a positive number, the remaining composition is favorable to the player.
To learn how to count cards using the Plus/Minus system is a three-step process:

1.) Turn over the cards in a deck, one at a time and state the value of the card. Don't try to keep a running count, just identify the value of each card. When you can do this without hesitating, it is time to go to step two.

2.) Turn over the cards, one at a time and keep a running total. Since the Plus/Minus system is a "balanced system", after you turn over the last card, your running count should be zero..

3.) Now you will begin to work on speed. Turn over the cards two at a time. If the total of the two cards adds to zero, then just drop them without saying anything. On a live game you will often see spreads of cards that add up to zero and you want to be able to develop the ability to recognize them. You will work up to the third exercise and do it until you can go through an entire deck in less than thirty seconds.

Converting a running count to a "true count."
A "true count" is the running count adjusted for the number of unseen decks and is the count the player will actually use to make decisions.

To convert a running count into a true count you merely *divide the running count by the total number of unseen decks.*

Example: The running count is +9 and there is three decks left. +9 / 3 = +3

Example: The running count is –6 and there is two decks left. –6 / 2 = -3.

As you are probably aware, whether the running count is a positive or negative number, converting it to a true count will never change the positive or negative aspect.

When there is less than one deck left unseen you can divide the running count by a fraction.

Example: The running count is +6 and there is 1/2 of a deck left. +6 / (1/2) = +12.

Example: The running count is –3 and there is 1/3 of a deck left. –3 / (1/3) = -9.
As long as the numerator of the fraction you are dividing by is "1", you merely have to *multiply* the running count by the denominator.
If the number of unseen decks is a whole number and a fraction, such as 2 1/2, I would just do the calculation twice and estimate the actual true count to be somewhere in the middle.

Example: The running count is +6 and there is 2 1/2 decks left.
+6 / 2 = +3 and +6 / 3 = +2 so I know the answer is greater than 2 and less than 3.
I have found the best way to practice my true count conversion is when I find myself in a section with the only live game being a shoe. I can then practice keeping a running count and converting it to a true count.

Learning the indices.
A card counter uses "indices" AKA "index numbers", "critical index numbers" or "matrix numbers" to tell him when to deviate from basic strategy. Learning the indices is absolutely necessary for the study of card counting. It is necessary for the card counter, since he won't be able to gain an advantage from the house without knowing when to deviate from basic strategy. It is necessary for the supervisor since, seeing a player make a decision that goes against basic strategy, can give him a false sense of security if indices are not known.

There are 140 indices for multi-deck games! The good news is there are only 18 that are considered the most important to learn and use and using them constitute 90% of the gain that can be achieved.

Multiple Deck Indices for the Plus/Minus Count

	2	3	4	5	6	7	8	9	T	A
16								+5	0	
15									+4	
14										
13	-1	-2								
12	+3	+2	0	-2	-3 (-1)					
11										+1
10									+4	+4
9	+1					+3				
TT				+5	+4					

Insurance – Take when the true count is +3 or higher.
TT = Two tens or picture cards.
Figure in parentheses is when dealer stands on all 17's.

Red – The player will stand when the true count is equal to or greater than the index number.
Blue – The player will double-down when the true count is equal to or greater than the index number.
Yellow – The player will split when the true count is equal to or greater than the index number.

HAND	AGAINST	ACTION	TC = or >
any	A	insure	3
16	10	s	0
15	10	s	4
10,10	5	split	5
10,10	6	split	4
10	10	d	4
12	3	s	2
12	2	s	3
11	A	d	1
9	2	d	1
10	A	d	4
9	7	d	3
16	9	s	5

HAND	AGAINST	ACTION	TC < or =
13	2	h	-1
12	4	h	0
12	5	h	-2
12	6	h	-1
13	3	h	-2

These indices will be used in approximately 20% of all hands.
Don’t forget that a true count of –3 is *less* than a true count of –2.
You might be wondering why the index number for 16 against a 10 is zero. After all, if the true count is zero shouldn’t basic strategy be followed? The reason is: basic strategy has already taken into consideration the values of the cards exposed to get the player in that position.

The minus 1 value for 13 against a 2 is another confusing aspect, as it would seem that it is suggesting that you disregard basic strategy and "hit" 13 against a 2 when the true count is –1 or less. Remember, all indices for the hands of 12 – 16 indicate when the player will *stand.* If the true count is less than the index number, the player will *hit*.

To make sense of indices it is important to remember that if the true count is a negative number it indicates that 10's and aces have been used. If the true count is positive, then it suggests that small cards have been used.

The first step to memorizing the indices is to confine yourself to memorizing the plays that could potentially have an index number assigned to it. That way, if you are looking at a player with 15 against a dealer 9, you know that you don't have to worry your pretty little head about what the index number is because there isn't one. After memorizing the relevant plays, you then start to memorize the appropriate values to go with them. As with basic strategy, it is helpful to print out blank forms for the indices and practice completing them by hand. You can also practice them on the job, even if you haven't been counting down the cards. For instance, if you see a player standing on 16 against a ten, you would be thinking that is only the correct play if the count is zero or higher.

And finally, don't even think about working with indices until you have mastered basic strategy and the count system, as indices will be useless until you have.

What it takes for a card counter to achieve a profit over the game.
There are three characteristics that can be used to identify the card counter from the average gambler:

- They must utilize a bet spread.
- They must deviate from basic strategy.
- They will only take insurance in high counts.

A player that leaves a table whenever the true count goes to minus, must still spread his bets to two units in order to gain a 1% advantage over the house. This player can expect to win about the equivalent to one betting unit per hour.
A player that sits through all counts and doesn't leave a table will have to spread his bets to four units in order to gain his 1% advantage and win his one unit bet per hour.

How to identify a counter and report your suspicions.
It is much easier to prove that a player isn't a card counter than to prove he is. It is virtually impossible to be sure a player is counting in under a half an hour. There are ways however, to prove that a player isn't counting. If a player makes a large bet it can be assumed that the deck has a positive count. If the player fails to follow basic strategy or disregards basic strategy in a situation where indices say it should only be disregarded in a minus count, we know this player isn't a counter.

Example: A player makes a large wager and hits a 10 and a 2 against a dealer 6. A true count of –3 would be required for the player to draw that card. Bottom line: either the player bet big into a minus count or he failed to make the correct play against a plus count. Therefore we can assume that he isn’t a counter.

Of course the possibility exists that a counter may intentionally make a bad basic strategy play when he has made a one-unit bet but he would have to be sure that you would notice it to make it worth risking even a small amount. Counters generally prefer to dissuade your suspicions by their acting performances rather than their play.

When it comes time to report your observations to your supervisor, you must be prepared to state the following facts:
What bet spread did the player use? Example: "He played one hand at $50 up to two hands at $200 each.
What was the count when he raised his bets? Example: "He bets $50 on minus counts and then $100 when the count hit +2 and then $200 each on two hands when the count is +4 or higher."

When did he deviate from basic strategy? Example: "He only hits 16 against a ten in a plus count, he split tens against a 5 when the count was +5 and he hit 13 against a 2 in minus counts."

How much do you calculate this player should win per hour? Example: "At his eight unit spread and the fact that he doesn’t change tables, I think he has a 1% advantage and should win just over one average bet per hour or about $125."